

Mahakumbh 2028 (Simhastha): Autonomous Crowd Management Platform

Submission Report

1. Executive Summary & Problem Statement

The Mahakumbh (Simhastha) in Ujjain is projected to draw over **39 million pilgrims** to the banks of the Shipra River over a single month. Managing a crowd of this magnitude manually is a logistical impossibility. Historically, traditional static crowd management—relying on radio communication and manual barrier deployment—has resulted in critical failures and tragic stampedes because ground commanders are reacting to disasters **after** they begin.

Our Solution: An autonomous, self-healing nervous system for the city.

This platform replaces reactive policing with predictive, mathematical automation. By ingesting live crowd density metrics and weather data, the system calculates real-time Safety Indexes for all major sectors (like Ram Ghat and the Mahakaleshwar Temple). If a sector reaches a critical threshold, our A* routing engine autonomously blacks out the dangerous roads and generates a safe, Google Maps-integrated diversion route before a stampede can occur.

2. Project Scope

This project was built to simulate and solve the exact logistical nightmares faced during mega-gatherings.

What the Software Currently Simulates:

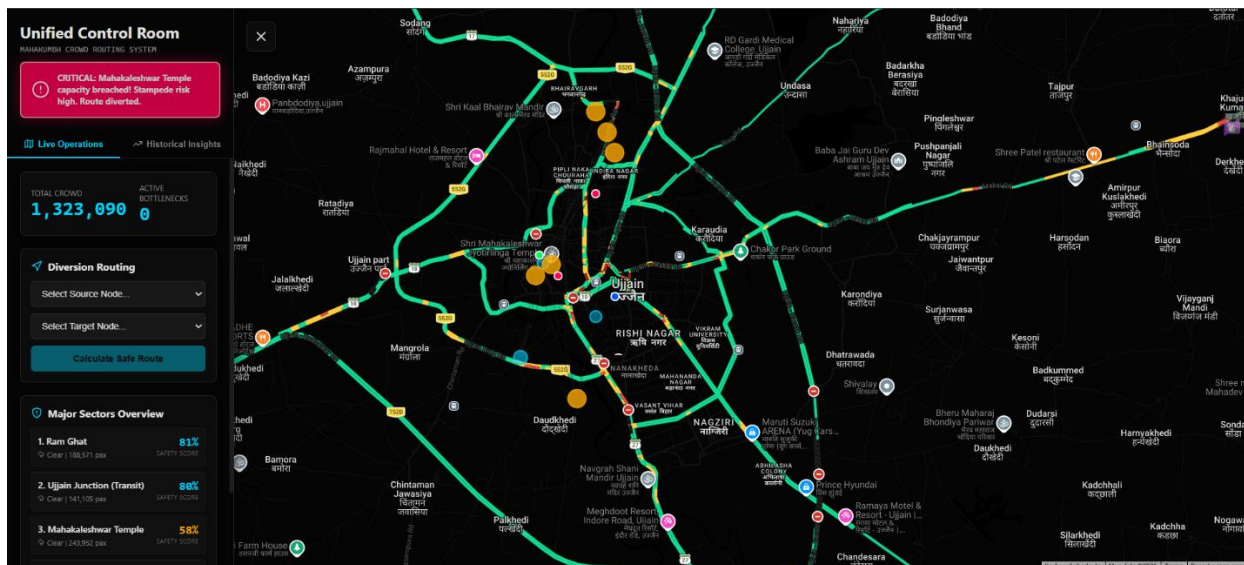
- **Live Traffic Loads:** High-frequency WebSocket streams representing rapid crowd fluctuations across 9 major nodes in Ujjain.
- **Dynamic Network Graphs:** A Euclidean distance-based graph of Ujjain where edge weights (road costs) actively change based on crowd density and weather penalties.
- **Predictive Modeling:** Machine Learning regression trained on historical Kumbh data to forecast the 39M+ capacity crunch.

The Goal: To prove that mathematical routing algorithms can operate faster than human decision-making during a panic scenario, preemptively neutralizing choke points.

3. Core Features & Functionality

A. Live Crowd Operations Dashboard

A centralized, highly responsive Glassmorphism interface acting as the "Unified Control Room." It provides a global view of total active crowds, ongoing bottlenecks, and a leaderboard of the most dangerous sectors sorted by their live **Safety Score**.



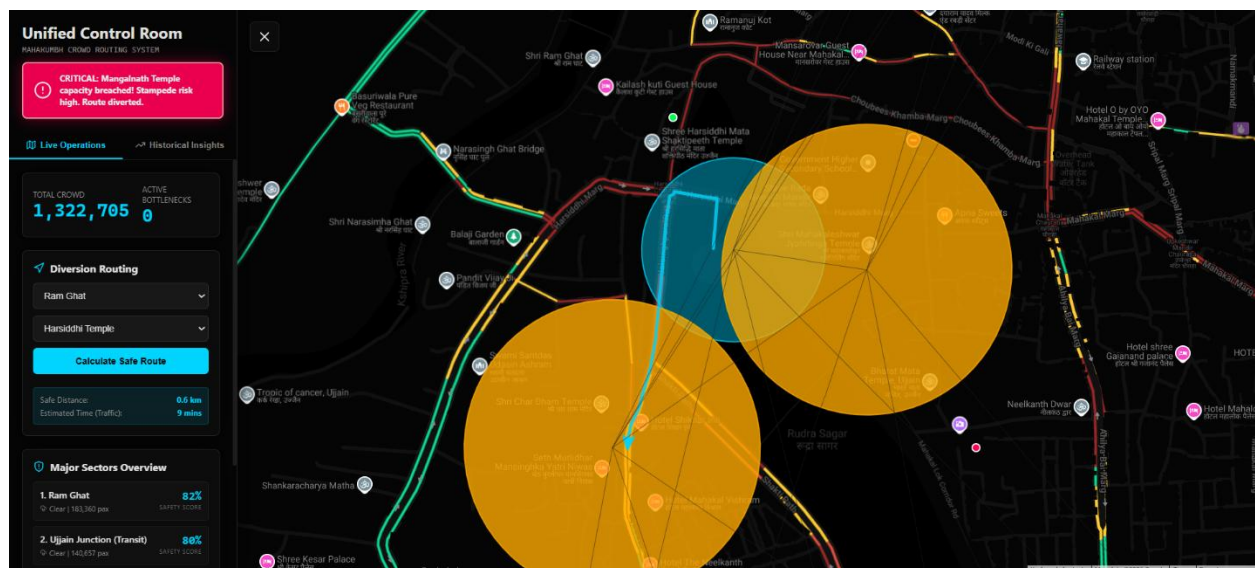
B. The Dynamic Safety Index Algorithm

Safety is not just about headcounts. Our backend algorithm dynamically calculates a score from 0% to 100% based on two live variables:

1. **Capacity Utilization:** Current crowd count vs. the physical square-meter limits of the Ghat.
2. **Weather Penalties:** Rain makes paths slippery and induces panic. Heavy rain dynamically subtracts up to 30% from the baseline safety score.

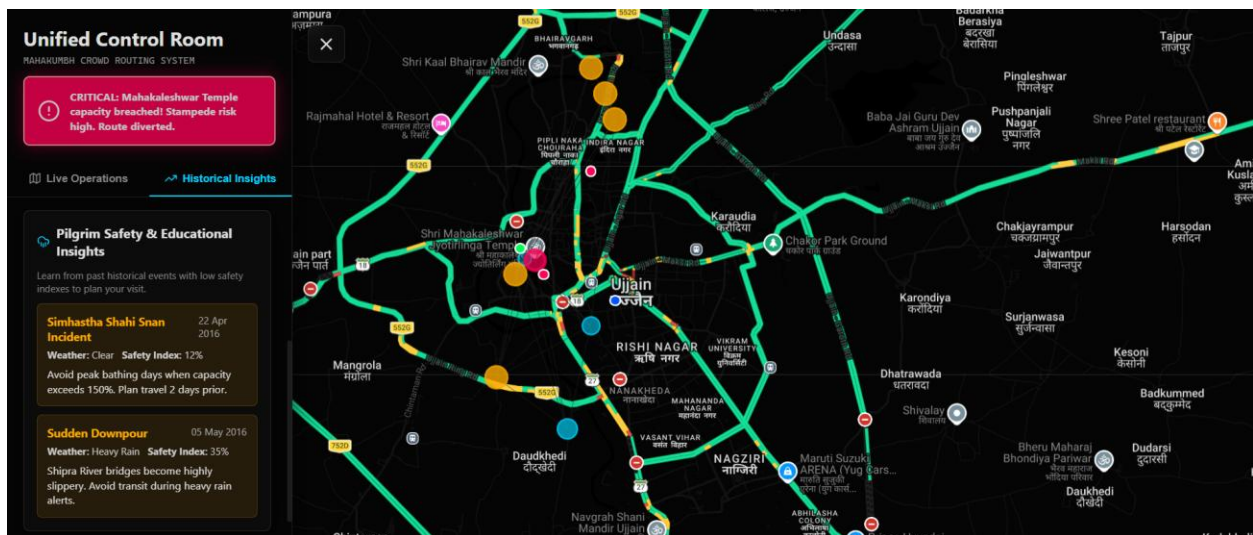
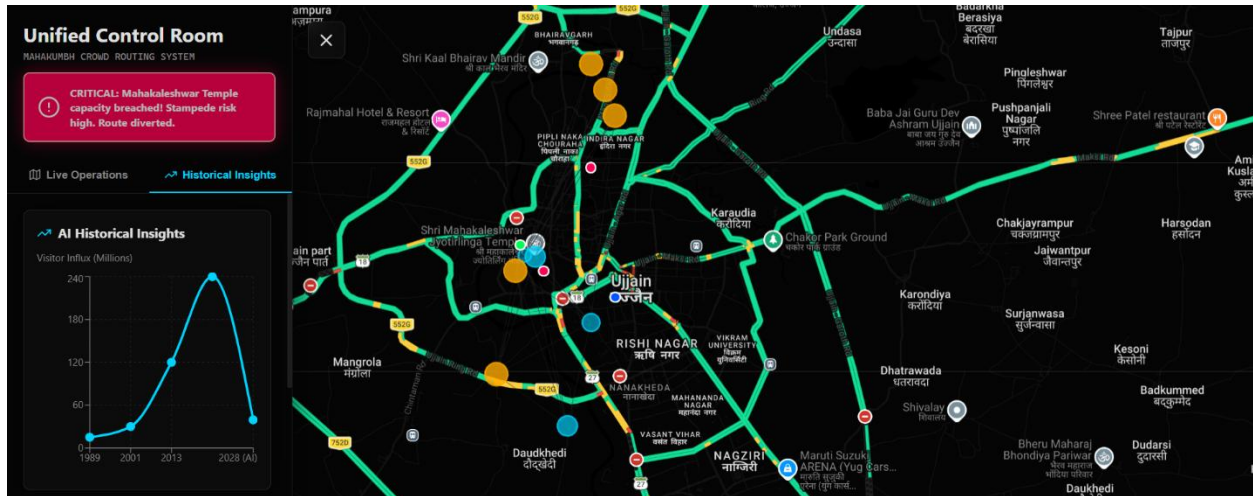
C. Autonomous Rerouting (The "Self-Healing" Network)

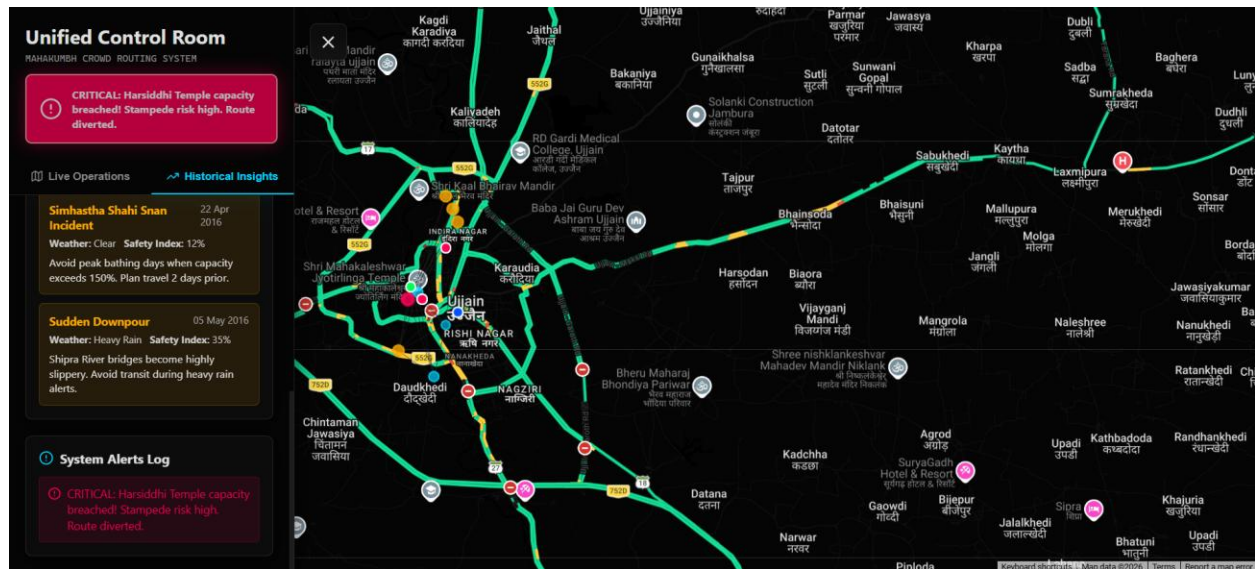
This is the platform's killer feature. When the simulator detects a sector's Safety Score dropping below 30%, it triggers a `CRITICAL` alert. The A* Search Algorithm immediately intercepts the graph, assigns an "infinite cost" to the dangerous sector, and autonomously calculates a brand-new diversion route to redirect incoming pilgrims safely away from the hazard.



D. Historical Insights & AI Projections

A dedicated module that utilizes a Scikit-Learn Machine Learning model to forecast the 2028 influx. It also features a rules-engine that analyzes past Simhastha incidents (e.g., the 2016 Shahi Snan) to provide actionable, automated safety advice to pilgrims and commanders.





4. Technical Architecture & Tech Stack

To handle city-scale traffic and mathematical computations with zero lag, the system is built on a highly concurrent, asynchronous architecture.

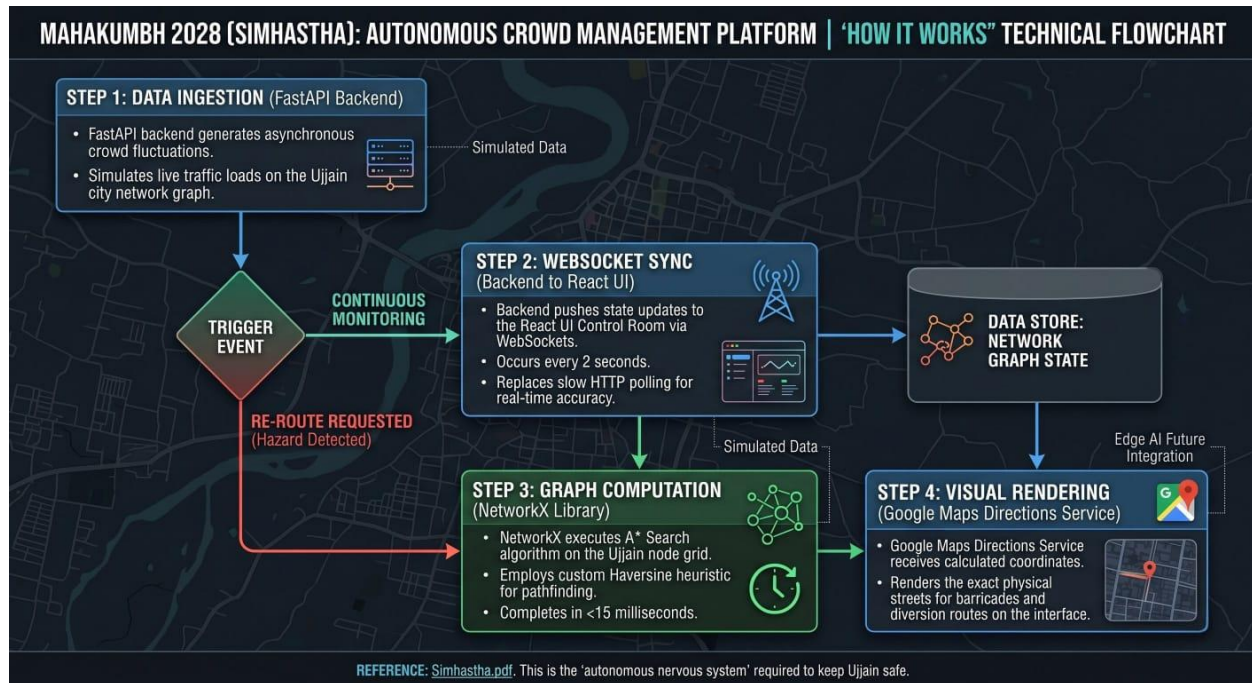
The Stack

- **Frontend:** Next.js 13, React 18, TypeScript, Custom CSS (Glassmorphism UI).
- **Backend:** Python 3.11, FastAPI (ASGI), SQLite, SQLAlchemy ORM.
- **Mapping:** Google Maps API (with custom dark-mode styling and physical road snapping).
- **Data Science:** Scikit-Learn, Pandas, NumPy, NetworkX.

How It Works (The Flow)

1. **Data Ingestion:** The FastAPI backend generates asynchronous crowd fluctuations.
2. **WebSocket Sync:** Instead of slow HTTP polling, the backend pushes state updates to the React UI via WebSockets every 2 seconds, ensuring military-grade real-time accuracy.
3. **Graph Computation:** When a reroute is requested, the **NetworkX** library executes an

A* Search across the Ujjain node grid. Because we use a custom Haversine heuristic, pathfinding across the city completes in under 15 milliseconds.



4. Visual Rendering: The calculated coordinates are sent to the Google Maps Directions Service, which beautifully renders the exact physical streets the police must barricade.

5. Further Steps: Real-World Edge AI Deployment

While the current software utilizes a highly accurate simulator to generate the WebSocket data, deploying this in Ujjain for 2028 requires physical hardware integration.

Current Hardware Architecture

Currently, the software runs on standard cloud infrastructure (e.g., Vercel and Render). However, a crowd of 39 million people will cause catastrophic failures in public 4G/5G networks due to bandwidth saturation.

Future Real-Life Deployment Requirements

To make this system a reality, we must transition from simulated data to physical Edge computing:

1. **Edge AI Vision Nodes:** Instead of streaming heavy 4K video to a cloud server, we will install **NVIDIA Jetson Nano** boards directly on CCTV poles at Ram Ghat and Mahakaleshwar. These Edge boards will run **YOLOv8** object detection locally to count heads, and only transmit tiny JSON payloads (e.g., `{"node": "ram_ghat", "count": 14500}`) back to our FastAPI server.

2. **LoRaWAN Radio Networks:** Because cellular networks will crash, the Edge cameras and the Control Room must communicate over dedicated, low-bandwidth LoRaWAN radio frequencies, ensuring the dashboard never goes offline.

3. **Automated Physical Response:** The routing engine's output will be linked to digital Variable Message Signs (VMS) on the highways and IoT-enabled smart gates that physically drop to block dangerous paths the millisecond our A* algorithm flags them as "infinite cost".

6. Conclusion

The 2028 Simhastha Mahakumbh presents a logistical challenge unparalleled in human history. Relying on human intuition to manage 39 million people is no longer viable. By combining highly concurrent WebSocket architecture, dynamic A* graph routing, and Edge AI capabilities, this platform proves that we can predict, mitigate, and physically reroute crowd disasters before a single life is endangered. We have not just built a dashboard; we have built the autonomous nervous system required to keep Ujjain safe.